# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A person has no pancreas. Which will be the most affected? 1

**(a)** absorption of oxygen **(b)** breathing
**(c)** digestion of all three food types **(d)** digestion of water only

**2.** If the ovaries do not function, a girl may not have: 1

**(a)** breathing **(b)** digestion **(c)** menstruation **(d)** growth

**3.** At the junction between the two breezes at around dusk, the wind: 1

**(a)** doubles **(b)** becomes a storm
**(c)** reverses at once strongly **(d)** nearly stops

**4.** A train leaves at 6 a.m. and runs at a steady 60 km/h. By 12 noon it has covered: 1

**(a)** 120 km **(b)** 60 km **(c)** 360 km **(d)** 720 km

**5.** A person breathes 16 times a minute, taking in 500 mL each time. The volume of air in one minute is: 1

**(a)** 4000 mL **(b)** 8000 mL **(c)** 16 mL **(d)** 500 mL

**6.** In which year is the average growth rate highest for girls? 1

**(a)** around 2 **(b)** around 11 to 13 years
**(c)** around 40 **(d)** around 20

**7.** Which of the following is a correct application of conduction? 1

**(a)** heating iron rod **(b)** sunlight **(c)** sea breeze **(d)** ceiling fan

**8.** A sprinter's reaction time is 0.2 s and a stopwatch is started after seeing the start. The measured 10 s may be: 1

**(a)** double **(b)** less than true by about 0.2 s
**(c)** exact **(d)** more than true

**9.** Herbivores have a longer small intestine than carnivores because: 1

**(a)** they sleep **(b)** plant food takes longer to digest
**(c)** they are slower **(d)** they are bigger

**10.** A student with tiredness, pale tongue and poor concentration has these symptoms mostly due to: 1

**(a)** scurvy **(b)** anaemia **(c)** rickets **(d)** goitre

**11.** A candle is placed under a paper spiral hung from a thread. The spiral rotates because: 1

**(a)** heat radiation pushes **(b)** cold air pushes it
**(c)** hot air rising pushes it **(d)** wind from the room

**12.** A man runs round a circular track of length 400 m in 100 s. His average speed is: 1

**(a)** 40 m/s **(b)** 400 m/s **(c)** 4 m/s **(d)** 0 m/s

**13.** **Assertion (A):** Digestion of starch begins in the mouth. 1
**Reason (R):** Saliva contains salivary amylase, which breaks starch into sugar.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** The pituitary gland is called the master gland. 1
**Reason (R):** The pituitary gland secretes the hormone insulin.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** The heat of the Sun reaches the Earth by radiation. 1
**Reason (R):** Conduction and convection need a material medium, which is almost absent between the Sun and the Earth.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Why do we place a cloth wet with water around a matka in summer? 2

**17.** Why can't the average speed of a round trip be the average of two speeds? 2

**18.** Explain why the right side of the heart has less thick walls than the left. 2

**19.** Why is emotional support as important as physical care during adolescence? 2

**20.** Why does ice wrapped in a blanket not melt for long? 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Explain why white blood cells are found in higher numbers at an infected wound. 3

**22.** Explain how adrenaline helps in exams or sports. 3

**23.** Explain why woollen blankets keep you warm but do not generate heat. 3

**24.** The speed of sound is 340 m/s. How long does the sound take to travel 1.7 km? 3

OR

Describe the role of fibre in preventing digestive disorders. 3

**25.** A teenager eats little fruit and many fried snacks and gets frequent mouth ulcers. Identify the likely deficiency and prescribe a diet. 3

OR

Explain how the design of a house with high ceilings, small windows and thick walls helps in hot regions. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** Sunita is cooking dal on a gas stove. She leaves a steel spoon in the pan and, after some time, the spoon's handle becomes too hot to hold. She uses a wooden spoon to stir. She also sees the dal circulating inside the pan as it boils. 5

(i) By which mode does the handle of the steel spoon get hot?
(ii) Why is the wooden spoon preferred?
(iii) By which mode does the dal circulate?
(iv) Explain how the water in a pan gets heated from the bottom to the top.

**27.** A car's odometer reads 25 340 km at 9:00 a.m. and 25 420 km at 11:00 a.m. Its speedometer reads 60 km/h at 9:30 a.m. and 20 km/h at 10:30. Explain the difference between average and instantaneous speeds. 5

OR

Explain how smoking and air pollution harm the respiratory, circulatory and excretory systems. 5

## END OF QUESTION PAPER Total: 50 Marks

MASTERMIND